

Exercise 6: Additional Worksheet

Practice Dataset

Execute K-Means with

$$K = 2$$

using Euclidean distance. Instead of random initialization, use the initial centroids

$$\mu_1^{(0)} = q_5 = (6.000, 2.000), \quad \mu_2^{(0)} = q_6 = (7.000, 3.000).$$

For each iteration, assign all points to the closest centroid, recompute the centroid positions, and stop when assignments no longer change.

Table 1: Short 6-point K-Means dataset

ID	x	y
q_1	1.000	7.000
q_2	2.000	8.000
q_3	3.000	7.000
q_4	5.000	1.000
q_5	6.000	2.000
q_6	7.000	3.000

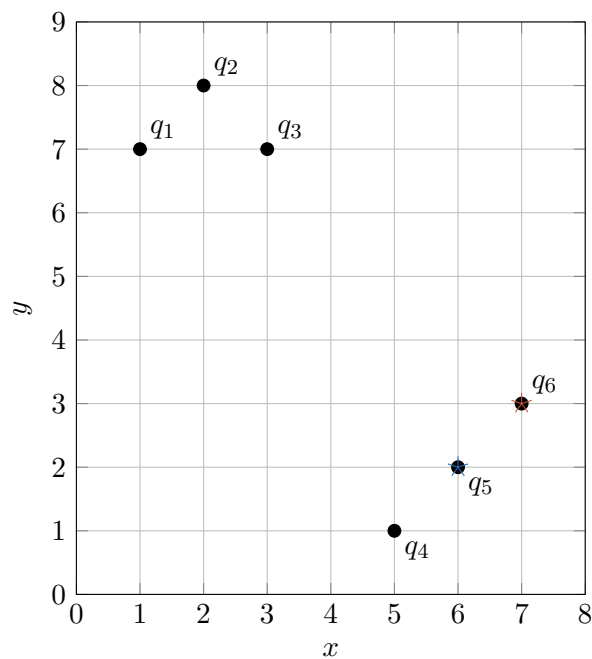


Figure 1: Dataset with the fixed initial centroids.